

Feolu Kolawole

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EDUCATION

Stanford University

Expected Graduation Date: June 2028

B.S. in Computer Science

- **Coursework:** Data Structures & Algorithms, Operating Systems, Concurrency, Linear Algebra, Multivariable Calculus, Machine Learning, Computer Vision, Reinforcement Learning, Statistics, Combinatorics
- **Activities/Societies:** Optiver FutureFocus (Quant), Stanford XR (VP), Stanford AI Club, Stanford Neurotech

PROFESSIONAL EXPERIENCE

Anthropic

Berkeley, CA

Research Fellow (Advised by Monte MacDiarmid)

Aug 2026 - Present

- Designing specialized **reinforcement learning** environments to study how post-training shapes model behavior.
- Applying **mechanistic interpretability** techniques to analyze learned representations and training dynamics.

Microsoft

Redmond, WA

Machine Learning Researcher (Advised by Dr. Olesya Melnichenko)

Jun 2026 - Aug 2026

- Trained a **computer vision** model that automates MRI scan calibration, cutting calibration overhead by **~40%**.
- Developed a **sequence file visualization** program that renders MRI undersampling patterns, enabling acquisition protocols to be inspected and iterated on before any physical scan is run.

Stanford Artificial Intelligence Laboratory

Stanford, CA

Machine Learning Researcher (PI - Fei-Fei Li & Ehsan Adeli)

Jan 2026 - Present

- Engineered a video-language pipeline to automatically monitor ICU patient behavior from continuous footage, enabling real-time health assessment of **2,656** critically ill patients without manual intervention.

Lead Machine Learning Researcher (PI - Ron Fedkiw)

Mar 2025 - Dec 2025

- Drove CNN models generating realistic 3D hair reconstructions across demographics, boosting accuracy to **96%**.
- Built a robust extraction pipeline using OpenCV and SAM, achieving perfect segmentation on **98%** of images.

Stanford Human Perception Lab

Palo Alto, CA

Machine Learning Researcher (PI - Khizer Khaderi)

Dec 2024 - May 2026

- Built a single model that predicts user actions from unlabeled video across **9 distinct environments**, achieving **85.2%** accuracy and enabling zero-shot action understanding without labeled training data.
- Engineered an AR application on Snap Spectacles in collaboration with the Snapchat Prototyping Team, enabling analysis of **92%** of the available field of vision.
- Created a process to transform video into accurate 3D environments by integrating SLAM and point-cloud segmentation models, thereby optimizing output by **44%**.

PUBLICATIONS

Cross-Game Semantic Alignment of Latent Action Representations ([IJCAI GLOW 2026 – Best Paper Candidate](#))

- Showed a single inverse dynamics model trained across racing, Atari, and first-person games clusters action embeddings by **action class rather than game identity**, with 50 labels per class recovering **91.1%** accuracy.

MoSV: Mixture-of-Steering Vectors for LLM Hallucination Mitigation ([ICML Mechanistic Interpretability 2026](#))

- Proposed a framework that reduces LLM hallucinations by dynamically selecting from multiple learned correction vectors per prompt, improving factual accuracy by **+2.4pp** where prior methods gained only **+0.3pp**.
- Demonstrated that the framework automatically discovers distinct types of hallucinations without manual labeling, evaluated across **10,615** items spanning **8 knowledge domains**.

PROJECTS

Power Lever: GPU-Efficient LLM Inference Gateway (Winner, Stanford Hackathon) | *Python*

- Built an inference gateway that dynamically routes LLM prompts to right-sized GPU hardware across **4 tiers**, reducing energy consumption by **75%** on simple queries while reserving high-end GPUs for complex tasks.

DYNAMO: Reinforcement Learning Portfolio Manager ([View Paper](#)) | *Python, Stable-Baselines3*

- Trained a reinforcement learning agent to automatically manage a portfolio across **10 asset classes**, achieving **15.76%** annualized returns with a **1.44 Sharpe ratio**, outperforming traditional strategies by **38–54%**.

Eous: Embodied AR Robot Assistant (Winner, Stanford Hackathon) | *Python, C#, Unity*

- Built a hands-free AR system integrating AR glasses, a smartphone, and a Raspberry Pi robot, enabling gesture and voice control with live camera feedback — all processed on-device with no cloud dependencies.

VR Healthcare Training Simulation (Winner, MIT Hackathon) | *Swift, RealityKit*

- Created a VR healthcare simulation on the Apple Vision Pro, enabling CPR and first aid training and pioneering SharePlay integration for collaborative learning with iPhone users.
- Showcased to **The Venture Reality Fund**, where the concept was acquired and carried forward.

Supreme Court Case Prediction (1st Place, Stanford Class of 500) ([View Paper](#)) | *Python, Numpy*

- Built a Bayesian court case prediction framework that achieved **73% top-3 accuracy** across **11 possible Supreme Court case outcomes** by leveraging observable case attributes.
- Featured as a winner and future example project in Stanford's CS109 course (selected from **500 students**).

SynchroSound: Facial Expression Based Song Selection | *Swift, API, Javascript, React, Objective-C*

- Built an iOS app that deciphers facial expressions to recommend mood-matching songs, integrating SwiftUI, UIKit, and SwiftData with Google Cloud Vision and Spotify's Web API.
- Processed **500+ test images** across multiple emotions to validate recommendations, improving user-song mood alignment **accuracy by 82%**.

Heap Allocator – Custom Computer Memory Management System | *C, x86 Assembly, GDB*

- Developed an algorithm that optimizes computer memory utilization (i.e. freeing up storage by **20%** across **100,000** requests).
- Implemented custom malloc, realloc, and free functions in C for heap-level memory management.

Shell - Custom Unix Shell Implementation | *C, C++, GDB, x86 Assembly*

- Built a command-line Unix Shell (computer operating system) that allows key tasks like file manipulation and redirection.
- Enhanced the Unix Shell to run **10+ programs** at once while testing across **100+ real-world** scenarios.

Medical MRI Image Reconstruction ([View Paper](#)) | *Python, Pytorch, Deep Learning*

- Engineered AI-based models for medical MRI image reconstruction with PyTorch, **cutting required scan times by 55–70%** while preserving quality images, and co-authoring a research paper on the work.
- Enhanced the MRI image quality beyond baseline deep learning models, **reducing reconstruction error by ~60%** and significantly improving detail preservation.

AgentTeX: Image to LaTeX (Winner, AgentOps Hackathon) | *Hugging Face, Python*

- Engineered an AI agent with OpenAI's Agents SDK to convert handwritten math to LaTeX with 91% accuracy.
- Featured as an AgentOps hackathon winner, with board member recommendation for commercial launch.

Music Recommendation System ([View Paper](#)) | *Python, SVD, Dimensionality Reduction, PCA*

- Designed a music recommendation system using SVD and PCA to analyze over **60 audio features** from **7,000 Spotify songs**, uncovering patterns in sound beyond traditional metadata.
- Implemented feature projection to identify similarities across **35 combinations of musical features**.

TECHNICAL SKILLS

Programming Languages: Python, Java, Javascript, Typescript, C, C#, C++, Swift, Assembly, HTML/CSS, R
Libraries/Frameworks: Next.js, React.js, Flask, FastAPI, Node.js, Tailwind CSS, Node.js, OpenGL, Git, Linux
Software: Github, Docker, Vim, Emacs, Pytorch, Visual Studio Code, XCode, Blender, Unity, Lens Studio, RStudio